

Next-generation Bayesian local robustness

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Berkeley Neyman Seminar

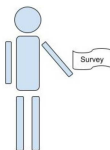
September 2026

The basic problem

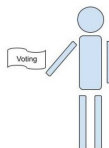
We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.



Observe (x_i, y_i) for $i = 1, \dots, N_S$



Observe x_j for $j = 1, \dots, N_T$

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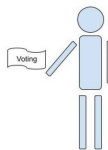
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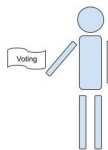
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This talk will study “Multilevel regression and poststratification” estimators:

- Set $\mathbb{E}[y|x, \theta] = m(\theta^\top x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\text{Survey data})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu}^{\text{MrP}}(Y_S) = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}[y|x_j]$.

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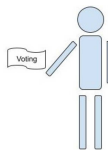
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Problem: MCMC is time consuming and the regression model is surely misspecified.

Goal: Meaningful regression specification checks with frequentist uncertainty.

Result preview

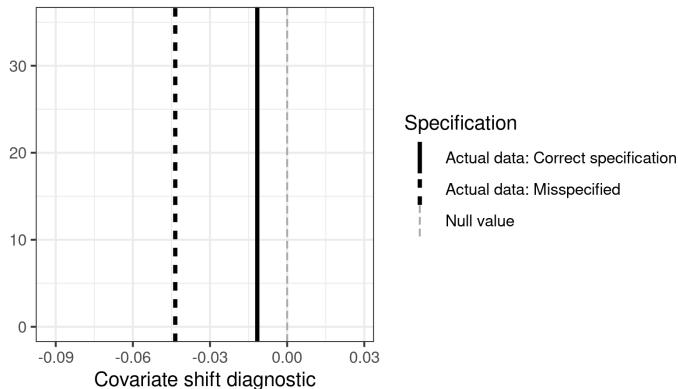


Figure 1: A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecified model is more extreme.

Problem: What is “large enough” to be real evidence of misspecification?

Possible answer: Frequentist confidence intervals via the parametric bootstrap.

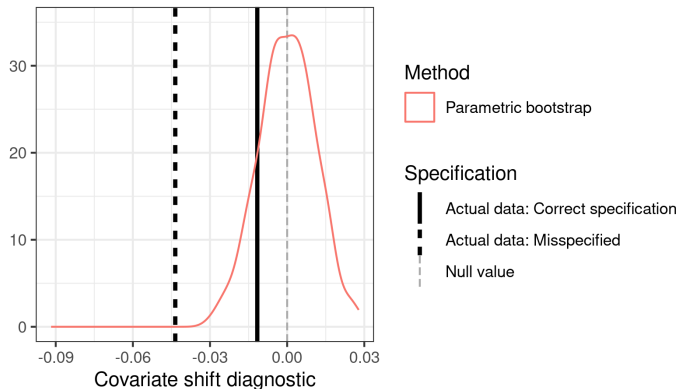


Figure 1: A covariate shift diagnostic

The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.

Possible answer: Infinitesimal jackknife confidence intervals? [Giordano and Broderick, 2026]

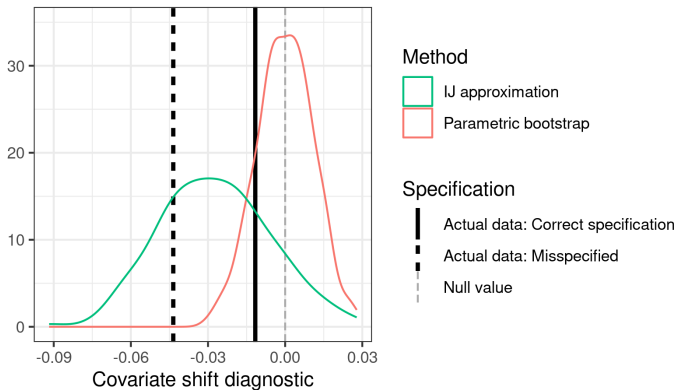


Figure 1: A covariate shift diagnostic

IJ confidence intervals are fast and often work well for posterior means.
But they are not accurate for this more complex diagnostic (high Monte Carlo error).

Result preview

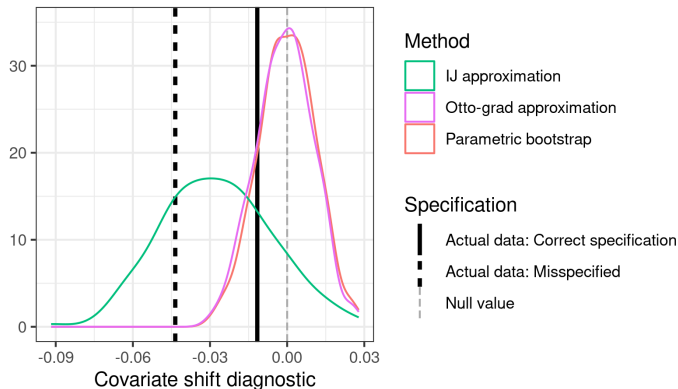


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But they are not accurate for this more complex diagnostic (high Monte Carlo error).

Answer: Our flow-based local sensitivity measures (“Otto grad”) produce good approximations to the bootstrap at negligible computational expense.

- Bayesian local regression specification checks
 - (With Alice Cima, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
 - (With Tamara Broderick)
- Flow-based Bayesian local robustness: “Otto grad”
 - (With Lucas Schwengber)

Bayesian local regression specification checks

Local specification checks

$$\hat{\mu}^{\text{MrP}}(Y_S) = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j \text{ where } \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})} [m(\theta^\top x_j)]}_{\text{Want to check without re-running MCMC}}.$$

This model of *survey data* is definitely misspecified.

Why do we care about correct specification? We want to use $\mathcal{P}_S(x)$ to learn about $\mathcal{P}_T(x)$.

Correct specification $\Rightarrow$ Regression estimates are invariant to $\mathcal{P}_S(x)$

Idea: Check invariance to $\mathcal{P}_S(x)$ directly.

Let $\mathcal{P}(x, y)$ denote a generic joint density of x, y .

Let $\hat{\mathcal{P}}_S$ the empirical distribution on the survey data.

1. Write our estimand as a functional $\mu(\mathcal{P})$ that takes in a density and returns a MrP estimate
2. Define a covariate density perturbation $w(x)$
3. Define $\mathcal{P}_\epsilon(x, y) = \mathcal{P}(x, y) + \epsilon w(x) \mathcal{P}(y|x) = (\mathcal{P}(x) + \epsilon w(x)) \mathcal{P}(y|x)$
4. Compute the derivative $\Delta(w, \mathcal{P}) = \partial \mu(\mathcal{P}_\epsilon) / \partial \epsilon$
5. Plug the empirical distribution into $\Delta(w, \hat{\mathcal{P}}_S)$ and check whether ≈ 0

Bayesian posterior functionals

We are interested in $\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})} [g(\theta)]$ where $g(\theta) := \frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j)$.

Let the regression log likelihood be $\ell(y|x, \theta)$.

Define the “generalized posterior” functional:

$$T(\mathcal{P}, N) := \frac{\int g(\theta) \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}{\int \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}.$$

Let $\hat{\mathcal{P}}_S$ denote the empirical distribution over x_i, y_i . Then

$$\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})} [g(\theta)] = \frac{\int g(\theta) \exp \left(N_S \frac{1}{N_S} \sum_{i=1}^{N_S} \ell(y_i|x_i, \theta) \right) \pi(\theta) d\theta}{\int \exp \left(N_S \frac{1}{N_S} \sum_{i=1}^{N_S} \ell(y_i|x_i, \theta) \right) \pi(\theta) d\theta} = T(\hat{\mathcal{P}}_S, N_S).$$

The directional derivative is

$$\left. \frac{\partial T(\mathcal{P}_\epsilon, N)}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta|\mathcal{P}, N)} \left(g(\theta), N \int \ell(y|x, \theta) w(dx) \mathcal{P}(y|x) dx dy \right)$$

Our diagnostic is then given by

$$\Delta(w, \hat{\mathcal{P}}_S) = \text{Cov}_{\mathcal{P}(\theta|\text{Survey data})} \left(\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j), \sum_{i=1}^{N_S} w(x_i) \ell(y_i|x_i, \theta) \right)$$

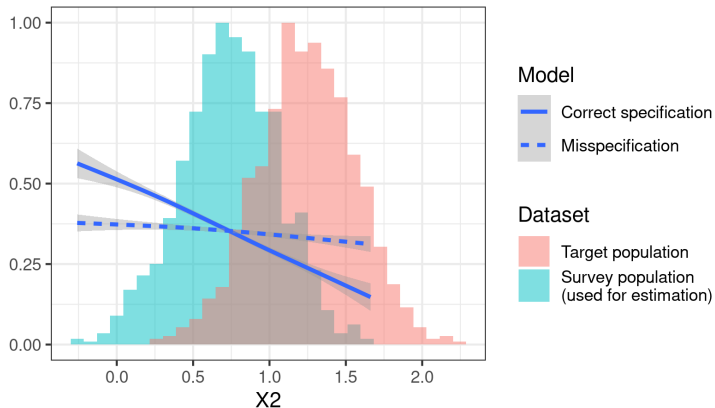


Figure 2: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

Model 0 (correct) : $y \sim \text{Logistic}(1 + X1 + X2 + X3 + X4 + X5)$

Model 1 (misspecified) : $y \sim \text{Logistic}(1 + X3 + X4 + X5)$

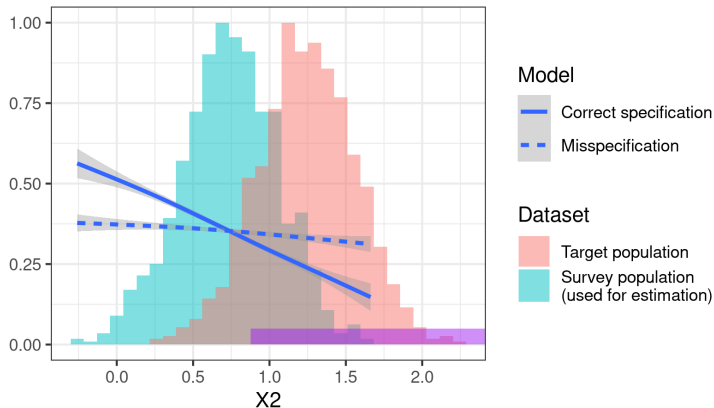


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Compute diagnostic with $w(x) = \mathbb{I}(X2 > 0.878)$

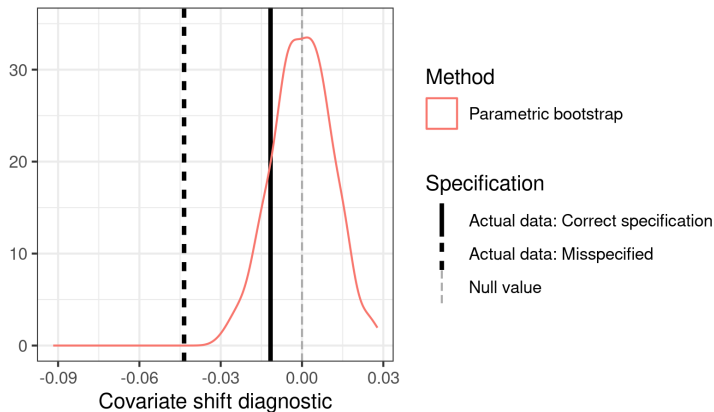


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Compute diagnostic with $w(x) = \mathbb{I}(X_2 > 0.878)$

Frequentist variance with the infinitesimal
jackknife

Why frequentist variance?

For the remainder of the talk, focus on approximating the *frequentist* sampling distribution under $x_i, y_i \stackrel{iid}{\sim} \mathcal{P}_S(x, y)$ of quantities like our diagnostic:

$$\Delta = \text{Cov}_{\mathcal{P}(\theta|\text{Survey data})} \left(\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j), \sum_{i=1}^{N_S} w(x_i) \ell(y_i | x_i, \theta) \right)$$

No need to commit to our covariate shift diagnostic to care!

- Frequentist variability of *any* model checking diagnostic is interesting
- Frequentist variability remains meaningful under model misspecification (Huggins and Miller, IJ papers)

We will form *local approximations* computable only from *the original posterior*.

We will compare with computationally expensive bootstrap:

- Compute Δ on the original dataset
- For $B = 1, \dots, 100$:
 1. Sample $(x_1^*, y_1^*), \dots, (x_{N_S}^*, y_{N_S}^*)$ (parametric or nonparametric bootstrap)
 2. Run MCMC to estimate $\mathcal{P}(\theta | (x_1^*, y_1^*), \dots, (x_{N_S}^*, y_{N_S}^*))$
 3. Use the draws to estimate Δ_B^*
- Compare Δ with quantiles of $\Delta_B^* - \Delta$

Bayesian von-Mises Expansion

Define the “generalized posterior” functional:

$$T(\mathcal{P}, N) := \frac{\int g(\theta) \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}{\int \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}.$$

Recall that $\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})} [g(\theta)] = T(\hat{\mathcal{P}}_S, N_S)$

To derive frequentist variance estimates, study the *von Mises expansion*.

Let $\mathcal{P}_t = t\hat{\mathcal{P}}_S + (1 - t)\mathcal{P}_S$. Then form the series expansion in t :

$$\begin{aligned} \sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})} [g(\theta)] - T(\mathcal{P}_S, N_S) \right) &= \sqrt{N_S} \left(T(\hat{\mathcal{P}}_S, N_S) - T(\mathcal{P}_S, N_S) \right) \\ &= \sqrt{N_S} \left. \frac{\partial T(\mathcal{P}_t, N_S)}{\partial t} \right|_{t=0} (1 - 0) + \mathcal{E}(\tilde{t}) \\ &= \underbrace{\sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi})}_{\text{Infinitesimal jackknife estimator}} + o_p(1) + \mathcal{E}(\tilde{t}). \end{aligned}$$

where

$$\psi_i := N_S \text{Cov}_{\mathcal{P}(\theta|\text{Survey data})} (g(\theta), \ell(y_i|x_i)) \quad \text{and} \quad \tilde{t} \in [0, 1].$$

Bayesian von-Mises Expansion Results

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})} [g(\theta)] - T(\mathcal{P}_S, N_S) \right) = \sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

- Assume (sketch): A well-behaved MAP *maximum a posteriori* estimator $\hat{\theta}$ exists.
 - The dimension of θ is fixed as $N \rightarrow \infty$
 - The expected log likelihood has a strict maximum at θ_0
 - The observed log likelihood satisfies $\hat{\theta} \rightarrow \theta_0$
 - The expected log likelihood Hessian is negative definite at θ_0
- Assume (sketch): We can apply standard asymptotics.
 - The log prior and log likelihood are four times continuously differentiable
 - The prior is proper, and a technical set of prior expectations are finite
 - The log likelihood and its derivatives are dominated by a square-integrable envelope function for all θ in a neighborhood of θ_∞ .

Theorem 3 [Giordano and Broderick, 2026] (sketch):

$$\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{P} 0 \quad \text{in the Bayesian von-Mises expansion.}$$

Corollary: Let $\frac{1}{N_S} \sum_{i=1}^{N_S} (\psi_i - \bar{\psi})^2 \xrightarrow{P} V$. By the CLT applied to ψ_i ,

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})} [g(\theta)] - T(\mathcal{P}_S, N_S) \right) \xrightarrow{d} \mathcal{N}(0, V).$$

R. Giordano and T. Broderick. The Bayesian infinitesimal jackknife for variance. *arXiv preprint arXiv:2305.06466*, 2026.